

LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Logistic Regression

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->Objective

The objective of this project is to predict whether a person has diabetes or not using Logistic Regression. The model classifies the outcome as 0 (No Diabetes) or 1 (Diabetes).

->Dataset

Dataset: Diabetes Prediction Dataset

Source: Kaggle

File: diabetes_cleaned.csv

The dataset contains information such as gender, age, hypertension, heart disease, smoking history, BMI, HbA1c level, and blood glucose level. The target variable is diabetes.

Model and Workflow

->Model: Balanced Logistic Regression

Workflow: Load dataset → inspect data → preprocess categorical data → train-test split → median imputation → StandardScaler → train Logistic Regression → evaluate model → interpret coefficients.

->Evaluation Metrics

Metric	Result
Accuracy	0.884711
Precision	0.425678
Recall	0.879717
F1-score	0.573736
ROC-AUC	0.960075

-> Confusion Matrix

TN: 15521

FP: 2013

FN: 204

TP:1492

-> Key Findings

The Logistic Regression model was able to classify diabetes outcomes using the given patient-related features. The evaluation metrics helped to understand the performance of the model. The confusion matrix showed the correct and incorrect predictions made by the model. The feature coefficients were also checked to understand which features had a stronger influence on the predictions.

->Conclusion

In this project, I used Logistic Regression to predict whether a person has diabetes or not. I prepared the dataset by handling missing values, converting categorical data into numerical form, and scaling the features. The model was then evaluated using accuracy, precision, recall, F1-score and ROC-AUC. This project helped me understand how Logistic Regression can be used for classification and how different evaluation metrics are used to check a machine learning model.